

8 Reaction kinetics

AS · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Rate can be measured from the tangent to concentration–time curves, gas collection or mass loss. Collision theory requires sufficient energy and suitable orientation; increasing temperature raises the fraction with energy above activation energy. Catalysts offer a lower activation energy pathway and do not alter equilibrium position or K .

Worked example: follow the reasoning

If [product] increases from 0.00 to 0.12 mol dm⁻³ in first 20 s, average rate is 0.0060 mol dm⁻³ s⁻¹.

Quick check · multiple choice

A What is the effect of a catalyst on activation energy? [1]

- A Raises it
- B Eliminates collisions
- C Changes ΔH
- D Lowers it

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Explain why a catalyst increases both forward and reverse reaction rates but does not change K_c . [3]

Answers and explanation

Quick check: D. Lowers it

Question 1 [3 marks]

It provides lower activation-energy route [1] for both directions [1]; equilibrium composition/thermodynamics at fixed T are unchanged [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Chemistry 9701, 2025–2027. <https://www.cambridgeinternational.org/Images/664563-2025-2027-syllabus.pdf>

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